

Sonifying the Heart: Inner Emotion Detection from ECG through Music

Music Workshop Sprint 3: Individual Contribution

Manidhar Sukasi 2023101067

1 My Role

I built the third stage of our pipeline, **Music**. I also put the whole pipeline together as one package and notebook.

2 What I Did

- **Melody.** One note per heartbeat.
 - Each note starts at the R-peak, lasts 85% of the R–R interval, and is as loud as the ECG at that beat.
 - The pitch follows the deviation d_i from the mean of the last five intervals. It is C4 on time, higher for a faster beat and lower for a slower one.
 - The melody stays on the C-major pentatonic scale while $|d_i| < 12\%$, moves chromatically beyond that, and is two octaves away at 50%.
- **Bug fixes.** Found and fixed two mapping bugs: faster beats went down instead of up, and downward steps left the scale.
- **Arrangement.**
 - Strings play C–Am–F–G, one chord per four beats. A chord turns diminished ($\geq 12\%$) or into a cluster ($\geq 25\%$) as the beats get irregular.
 - A soft “lub-dub” bass-drum pulse follows every beat.
 - The melody instrument follows the detected emotion.
- **Rendering.**
 - MIDI with `pretty_midi`, and audio with FluidSynth and the FluidR3 soundfont, normalised to 85% peak.
 - A score plot that lines up the ECG, the beat deviations and the notes.
- **Integration.** The `ecgmusic` package with its command line (`python -m ecgmusic`), the `main.ipynb` walkthrough, and the repository documentation.

3 Key Result

- **A steady heart sounds calm.** All normal pieces stay on the scale, and all 67 of their chords are consonant. Three melodies hold C4 as a pedal tone under the moving chords.
- **An irregular heart sounds restless.** AFib pieces leave the scale for 31–66% of their notes, and 68 of their 111 chords turn tense or chaotic.

4 Next Steps

- Regenerate all results with the package.
- Tune the thresholds by ear.
- Run a listening study with classmates.